Program: **Diploma in Microelectronics**	
Course Code: **3499**	Course Title: **Simulation Lab 1**
Semester: **3**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- To simulate and test mathematical and functional aspects of electronics, analog and digital electronic circuits using the available numerical computing software. (eg: MATLAB/SIMULINK, SCILAB/XCOS, LTSPICE, GNU OCTAVE)

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic mathematics	1002 2002	Mathematics I, II	1&2
Characteristics of basic electronic circuits	2041 3043	Basic Electronics Electronic circuits	2 3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Develop programs to perform basic matrix operations, solve equations and plot basic signals	12	Applying
CO2	Develop simulations to study the characteristics of electronic devices	6	Applying
CO3	Develop simulations to demonstrate basic electronic circuits and study their characteristics	9	Applying
CO4	Develop simulation models to implement low frequency and high frequency oscillator circuits.	12	Applying
	Lab Exam	6	Applying

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3		3			
CO2	3	3	3	3			
CO3	3		3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Develop programs to perform basic matrix operations, solve equations and plot basic signals.**		
M1.01	Perform basic matrix operations - addition, subtraction, multiplication, division and inverse (Verify mathematically)	3	Applying
M1.02	Solution of linear constant coefficient differential equation (Verify mathematically)	2	Applying
M1.03	Determine roots of a polynomial (Verify mathematically)	2	Applying
M1.04	Generate sine wave and cosine wave and plot in same window	2	Applying
M1.05	Plot different types of plots 3D, surface plot, polar plot	3	Applying
CO2	**Develop simulations to study the characteristics of electronic devices**		
M2.01	Plot input-output characteristics of diode	3	Applying
M2.02	Plot input-output characteristics of NPN BJT & MOSFET	3	Applying
	Lab Exam 1	3	
CO3	**Develop simulations to demonstrate basic electronic circuits and study their characteristics**		

M3.01	Design a single stage CE amplifier with potential barrier divider bias and 1) Plot its frequency response and 2) Measure the Bandwidth	3	Applying
M3.02	Build an emitter follower circuit and 1) Observe the frequency response 2) Measure the gain 3) Plot its input/output waveform	3	Applying
M3.03	Construct a single stage tuned amplifier circuit and 1) Plot input/output waveform 2) Plot its frequency response	3	Applying
CO4	**Develop simulation models to implement low frequency and high frequency oscillator circuits**		
M4.01	Design a RC phase shift oscillator and 1) Plot the output waveform 2) Measure the frequency of oscillation	3	Applying
M4.02	Design a Weinbridge oscillator and 1) Plot the output waveform 2) Measure the frequency of oscillation	3	Applying
M4.03	Design a Hartley oscillator and 1) Plot the output waveform 2) Measure the frequency of oscillation	3	Applying
M4.04	Design a Colpitts oscillator and 1) Plot the output waveform 2) Measure the frequency of oscillation	3	Applying
	Lab Exam 2	3	

Student Activity

Suggested Open-ended Experiments:

Students can do open ended experiments as a group of 2-3. There is no duplication allowed in experiments in between groups. This is mainly for the purpose of continuous internal evaluation. Students should include the open ended experiment in their lab record.

Examples :

1. Cascade amplifier
2. Power amplifier
3. Multivibrator circuits etc.

Text/Reference:

T/R	Book Title/Author
T1	Agam kumar Tyagi, MATLAB and Simulink for Engineers, Oxford Higher Education,2011
T2	Anil Kumar Verma, Scilab A Beginner's Approach, Cengage India, 2021
R1	Cleve Moler, Experiments with MATLAB, 2011
R2	Sandeep Nagar, Introduction to Scilab : For Engineers and Scientists, Kindle Edition, Apress, 2017

Online Resources:

Sl. No.	Website Link
1	https://www.mathworks.com/content/dam/mathworks/mathworks-dotcom/moler/exm/book.pdf
2	https://www.mccormick.northwestern.edu.documents/students/undergraduate/introduction-to-matlab.pdf
3	https://www.scilab.org
4	https://scilab.in/spoken-tutorial
5	https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html#learn